

The Anderson impurity model with a discrete bath

The spectral machinery of

F. B. Kugler, S.-S. B. Lee and J. von Delft, *Multipoint Correlation Functions: Spectral Representation and Numerical Evaluation*, [Phys. Rev. X **11**, 041006 \(2021\)](#)

applied to the model of the paper's Sec. IV: the Anderson impurity model, Eq. (79), with the band replaced by $N_b \leq 3$ bath sites so that exact diagonalisation is possible ($4^{N_b+1} = 16, 64, 256$ states). Nothing downstream of `spectrum(m)` is new. The PSFs of Eq. (28), the kernels of Eqs. (46), (67b), the PSF-level subtraction of Eq. (31) and the Keldysh amputation with the Eq. (D2) legs are the ones validated on the atom and the dimer.

$$H = -\frac{U}{2} \sum_{\sigma} n_{d\sigma} + U n_{d\uparrow} n_{d\downarrow} + \sum_{n=1}^{N_b} \sum_{\sigma} \left[\varepsilon_n b_{n\sigma}^{\dagger} b_{n\sigma} + V_n (d_{\sigma}^{\dagger} b_{n\sigma} + b_{n\sigma}^{\dagger} d_{\sigma}) \right],$$
$$\Delta(i\nu) = \sum_n \frac{|V_n|^2}{i\nu - \varepsilon_n}.$$

Provenance. The task sheet's closed forms (moment matching, the $N_b = 1$ spectrum, the discrete-bath χ_0) and its reference numbers come from the sheet author's independent Python exact diagonalisation. They are checked here, not assumed. Paper equations were read from the printed page: Eq. (79), and Eqs. (83)–(84) with "its particle-particle counterpart yields $-\chi_0$ ", on p. 16.

Item	Check	Result
Sec. 1	sum rule, $\varepsilon_n \rightarrow -\varepsilon_{N_b+1-n}$, anticommutators, $\langle n_{d\sigma} \rangle = \frac{1}{2}$	exact / $\leq 1\text{e-}13$ / $\leq 1\text{e-}12$
I1	chain-walk PSFs = dense PSFs, peak by peak	identical to rounding
I2	Dict peak merging = pairwise merging; no straddled cells	identical
I3	<code>ImpurityModel</code> interface, <code>oto1</code> threaded everywhere	existing 58,708 assertions unchanged
A1	$N_b = 1$ levels, Z , 8 poles, $T \rightarrow 0$ residues	$\leq 1\text{e-}13$; 0.172914 / 0.327086
A2	$G_d(i\pi T)$, 12 reference values	$\leq 4\text{e-}14$ relative
A3	$U = 0$: $[z - \Delta(z)]^{-1}$, MF and KF	MF $\leq 4\text{e-}14$; KF $\leq 4\text{e-}11$ on values up to $1/\gamma_0$
B1	$U = 0$ null test, MF and 16 KF components	$S^{\text{con}} \leq 2 \times 10^{-15}$, $F \equiv 0$

Item	Check	Result
B2	$V \rightarrow 0: F \rightarrow F^{\text{atom}}$	exact at $V = 0$; deviation $\propto V^2$
B3	weak coupling, Eq. (84) with the discrete-bath χ_0	$\leq 1.3\text{e-}5$ (3-point Richardson)
B4	crossing, both forms	$\leq 4\text{e-}13$
B5	$G^{1111} = 0, F^{2222} \equiv 0, F^{[\eta]}$ vs regular Matsubara objects	exact / exact / $\leq 7\text{e-}13$
B6	cost of tables and grids, $N_b = 1, 2$	see B6
O1	χ_{loc} table	all digits
O2	$\delta(\omega_{12})$ term vs $\beta\Delta_{\text{ST}}$	cut off, but at the doublet gap $a - b$, not Δ_{ST}
O3	heat maps next to the atom's	figures below

Running. The whole notebook takes about 15 minutes with 4 threads (`JULIA_NUM_THREADS=4`); almost all of it is the $N_b = 2$ Keldysh slices. On Colab, upload all of `src/` (now including `aim.jl` and `psf_chain.jl`), keep the folder layout described in the atom notebook, and `Pkg.add(["StaticArrays", "CairoMakie", "JLD2", "BenchmarkTools"])` first.

```
In [1]: include(joinpath(@__DIR__, "..", "src", "HubbardAtom.jl"))
using .HubbardAtom
using LinearAlgebra, Printf, BenchmarkTools, JLD2, CairoMakie
CairoMakie.activate!(type = "png", px_per_unit = 2)
FIGDIR = joinpath(@__DIR__, "figures", "aim"); mkpath(FIGDIR)
imp4(m, ops, s) = (impurity_operators(m, ops, :up)..., impurity_operators(m, ops, s === :updn ? :dn : :up)...)
# the MF vertex from the tables at arbitrary integer Matsubara indices
function Fmf(m, tabs, ms)
    Gc = mf_correlator_point(ms, m.β, tabs.con)
    Gc / (mf_leg(ms[1], m.β, tabs.leg) * mf_leg(-ms[2], m.β, tabs.leg) * mf_leg(ms[3], m.β, tabs.leg) * mf_leg(-ms[4], m.β, tabs.leg))
end
FREQS = [(1, 3, 5, -9), (1, -1, 3, -3), (1, 3, -1, -3), (1, 3, -3, -1), (1, -1, 1, -1), (3, -1, -3, 1), (5, -3, 1, -3)]
@printf("Julia %s, %d thread(s)\n", VERSION, Threads.nthreads())
```

Julia 1.13.0, 4 thread(s)

Sec. 1 — model and bath discretisation

Moment matching on a partition $x_0 < \dots < x_{N_b}$ of $[-D, D]$ [derived on the sheet]: $|V_n|^2 = \frac{\Delta}{\pi}(x_n - x_{n-1})$, $\varepsilon_n = \frac{1}{2}(x_{n-1} + x_n)$, so that $\sum_n |V_n|^2 = 2D\Delta/\pi$ for any partition. The presets use $D = 1$. The logarithmic partition $\{-D, -D/\Lambda, D/\Lambda, D\}$ is available through `box_edges(3;  $\Lambda$ )`, but every benchmark uses the presets.

The operators come from `jordan_wigner(2(Nb+1))` in mode order $d_\uparrow, d_\downarrow, b_{1\uparrow}, b_{1\downarrow}, \dots$. The hybridisation $d_\sigma^\dagger b_{n\sigma}$ crosses every mode in between

and gets those string signs from the matrix products. Degenerate blocks are gauge-fixed with $N + \sqrt{2}S_z$.

```
In [2]: println(" Nb edges                      ε_n                      |V_
n|^2·π/Δ          Σ|V|^2 vs 2Δ/π")
for Nb in 1:3
    ε, V = discretize_box(0.1, 1.0, box_edges(Nb))
    @printf(" %d %-28s %-26s %-15s %.2e\n", Nb, string(round.(box_edges
(Nb); digits = 3)), string(round.(ε; digits = 3)),
            string(round.(abs2.(V) .* π ./ 0.1; digits = 3)), abs(sum(abs
2, V) - 0.2 / π))
end
for Nb in 1:3
    m = aim_model(Nb; U = 1.3, β = 2.0); ops = operators(m); n = size(op
s.H, 1); Id = Matrix{1.0I, n, n}
    car = maximum(max(norm(ops.c[a] * ops.c[b]' + ops.c[b]' * ops.c[a] -
(a == b) * Id), norm(ops.c[a] * ops.c[b] + ops.c[b] * ops.c[a]))
                for a in eachindex(ops.c), b in eachindex(ops.c))
    nd = maximum(abs(thermal_average(spectrum(aim_model(Nb; U = 1.3, β =
β)), (impurity_operators(m, ops, σ)[2] * impurity_operators(m, ops, σ)
[1])) - 0.5)
                for β in (0.5, 10.0, 100.0), σ in (:up, :dn))
    @printf("Nb=%d: %3d states, %2d modes: max anticommutator deviation
%.1e, max |(n_dσ) - ½| over β ∈ {0.5, 10, 100} = %.1e\n",
            Nb, n, length(ops.c), car, nd)
end
```

Nb	edges	ε_n	V_n ^2·π/Δ	Σ V ^2 vs 2Δ/π
1	[-1.0, 1.0]	[0.0]	[2.0]	0.00e+00
2	[-1.0, 0.0, 1.0]	[-0.5, 0.5]	[1.0, 1.0]	1.39e-17
3	[-1.0, -0.333, 0.333, 1.0]	[-0.667, 0.0, 0.667]	[0.667, 0.667, 0.667]	1.39e-17

Nb=1: 16 states, 4 modes: max anticommutator deviation 0.0e+00, max |(n_dσ) - ½| over β ∈ {0.5, 10, 100} = 5.6e-16
Nb=2: 64 states, 6 modes: max anticommutator deviation 0.0e+00, max |(n_dσ) - ½| over β ∈ {0.5, 10, 100} = 5.3e-14
Nb=3: 256 states, 8 modes: max anticommutator deviation 0.0e+00, max |(n_dσ) - ½| over β ∈ {0.5, 10, 100} = 1.0e-15

I3 — an impurity-model interface

`propagator`, `correlator_4p`, `disconnected_4p`, `connected_4p`, `vertex` and `heatmap_tables` now take any `m::ImpurityModel` (`HubbardAtomModel`, `AIMModel`). They obtain $d_\sigma, d_\sigma^\dagger$ from `impurity_operators(m, ops, σ)`. Eq. (73) holds unchanged, because $\langle d_\sigma d_{\sigma'} \rangle = 0$ by charge conservation. `propagator_exact`, `vertex_exact` stay atom-only. `otol` now reaches `permuted_psfs` on every path, with a model-dependent default (`default_otol`: 0 for the atom, `1e-12` for the AIM, where H is not diagonal in the Fock basis). The existing 58,708 assertions pass unchanged.

A1 — closed-form spectrum for $N_b = 1$ [derived on the sheet]

With the bath site at $\varepsilon = 0$, $a \equiv \frac{1}{4}\sqrt{U^2 + 64V^2}$ and $b \equiv \frac{1}{4}\sqrt{U^2 + 16V^2}$, the levels are $-U/4 - a$ (1, singlet ground state), $-U/4 - b$ (4), $-U/2$ (3,

triplet), 0 (3), $-U/4 + b$ (4) and $-U/4 + a$ (1), with
 $Z = 3 + 3e^{\beta U/2} + 8e^{\beta U/4} \cosh \beta b + 2e^{\beta U/4} \cosh \beta a$. G_d has eight poles, at
 $\pm(a+b), \pm(a-b), \pm(U/4+b), \pm(U/4-b)$. They are read off the $\ell = 2$
PSFs below.

```
In [3]: for (U, β, Δ) in ((2.0, 10.0, 0.1), (0.5, 3.0, 0.1), (3.0, 1.0, 0.4))
        m = aim_model(1; U = U, β = β, Δ = Δ); sp = spectrum(m)
        lev = sort(vcat([fill(e, d) for (e, d) in aim_levels_nbl(m)]...))
        P = propagator_poles(m; sp = sp)
        @printf("(U, β, Δ) = (%.1f, %.4f, %.1f): levels %.1e, Z %.1e, %d pol
es, positions %.1e, Σ residues - 1 = %.1e\n", U, β, Δ,
                maximum(abs, sp.E - lev), abs(sp.Z / aim_partition_function_n
b1(m) - 1), length(P),
                maximum(abs, [p[1] for p in P] - aim_poles_nbl(m)), sum(p[2]
for p in P) - 1)
        end
        m = aim_model(1; U = 2.0, β = 1000.0)
        println("\nT → 0 (β = 1000), (U, Δ, D) = (2, 0.1, 1): residues of G_d")
        for (E, r) in propagator_poles(m)
            @printf("    pole %.6f    residue %.6f\n", E, r)
        end
        println("sheet: 0.172914 at each of ±(a-b), 0.327086 at each of ±(a+b)")
```

```
(U, β, Δ) = (2.0, 10.0, 0.1): levels 1.1e-15, Z 7.1e-15, 8 poles, positions 1.1e-15, Σ residues - 1 =
2.2e-16
(U, β, Δ) = (0.5, 3.0, 0.1): levels 7.8e-16, Z 1.3e-15, 8 poles, positions 6.7e-16, Σ residues - 1 =
2.2e-16
(U, β, Δ) = (3.0, 1.0, 0.4): levels 3.2e-15, Z 0.0e+00, 8 poles, positions 3.0e-15, Σ residues - 1 =
2.2e-16
```

```
T → 0 (β = 1000), (U, Δ, D) = (2, 0.1, 1): residues of G_d
pole -1.270441    residue 0.327086
pole -1.060055    residue 0.000000
pole -0.150330    residue 0.172914
pole -0.060055    residue 0.000000
pole +0.060055    residue 0.000000
pole +0.150330    residue 0.172914
pole +1.060055    residue 0.000000
pole +1.270441    residue 0.327086
sheet: 0.172914 at each of ±(a-b), 0.327086 at each of ±(a+b)
```

A2 — reference values of $G_d(i\pi T)$

$D = 1$, $\Delta = 0.1$, the presets of Sec. 1. The sheet's values come from its
independent Python ED. The target is 10^{-12} relative, with $|\operatorname{Re} G_d|$ at rounding
level (particle-hole symmetry). $N_b = 3$ runs through the chain walk.

```
In [4]: ref = [(1, 0.5, 10, -1.768830080892028), (1, 0.5, 100, -0.48514000345309
7), (1, 2.0, 10, -0.737281826717041),
                (1, 2.0, 100, -0.473355579919950), (2, 0.5, 10, -2.03069173304682
1), (2, 0.5, 100, -0.949216078082626),
                (2, 2.0, 10, -0.429277187551225), (2, 2.0, 100, -0.05982003883075
4), (3, 0.5, 10, -2.000856350026648),
                (3, 0.5, 100, -1.391281201028393), (3, 2.0, 10, -0.51373826406429
1), (3, 2.0, 100, -1.162698345747195)]
        println(" Nb    U    β    Im G_d(iπT)          sheet          r
el. diff    |Re G_d|    PSF method")
        for (Nb, U, β, r) in ref
            m = aim_model(Nb; U = U, β = β); sp = spectrum(m)
            G = propagator(m, fermionic_freq(0); sp = sp)
```

```
@printf(" %d %.1f %4d %.15f %.15f %.1e %.1e %s\n", Nb,
U,  $\beta$ , imag(G), r, abs(imag(G) - r) / abs(r), abs(real(G)),
default_psf_method(sp))
end
```

Nb	U	β	Im G_d(inT)	sheet	rel. diff	Re G_d	PSF method
1	0.5	10	-1.768830080892031	-1.768830080892028	1.8e-15	1.6e-15	dense
1	0.5	100	-0.485140003453099	-0.485140003453097	5.0e-15	5.8e-15	dense
1	2.0	10	-0.737281826717042	-0.737281826717041	1.1e-15	4.2e-17	dense
1	2.0	100	-0.473355579919952	-0.473355579919950	5.3e-15	7.7e-15	dense
2	0.5	10	-2.030691733046817	-2.030691733046821	1.7e-15	1.9e-15	chain
2	0.5	100	-0.949216078082621	-0.949216078082626	4.9e-15	3.9e-14	chain
2	2.0	10	-0.429277187551225	-0.429277187551225	6.5e-16	1.2e-15	chain
2	2.0	100	-0.059820038830754	-0.059820038830754	8.1e-16	2.4e-14	chain
3	0.5	10	-2.000856350026630	-2.000856350026648	9.1e-15	1.8e-14	chain
3	0.5	100	-1.391281201028346	-1.391281201028393	3.4e-14	1.4e-13	chain
3	2.0	10	-0.513738264064288	-0.513738264064291	5.0e-15	2.8e-15	chain
3	2.0	100	-1.162698345747233	-1.162698345747195	3.2e-14	2.3e-14	chain

A3 — the non-interacting limit

At $U = 0$: $G_d(i\nu) = [i\nu - \Delta(i\nu)]^{-1}$ exactly. In the KF, G_d^R from Eq. (67) at finite γ_0 equals $[z - \Delta(z)]^{-1}$ at $z = \omega + i\gamma_0$ exactly, since both are the same rational function.

```
In [5]: for Nb in 1:3
        m = aim_model(Nb; U = 0.0,  $\beta$  = 10.0); sp = spectrum(m); Os = impurity
        _operators(m, operators(m), :up)
        emf = maximum(abs(propagator(m, fermionic_freq(k); sp = sp) - propaga
        tor_free(m, im * value(fermionic_freq(k), 10.0))) for k in -4:3)
        cache = permuted_psfes(sp, Os; otol = 1e-12)
        ekf = maximum(abs(keldysh_correlator(m, Os, [ $\omega$ , - $\omega$ ], [1];  $\gamma_0 = \gamma$ , sp
        = sp, cache = cache) - propagator_free(m,  $\omega + im * \gamma$ ))
        for  $\omega$  in (-1.3, -0.4, 0.0, 0.25, 0.9),  $\gamma$  in (0.1, 0.0
        1))
        @printf("Nb=%d: MF, 8 Matsubara points: max |G_d - [i $\nu$  -  $\Delta(i\nu)]^{-1}$ | =
        %.1e; KF G^R, 10 points: %.1e\n", Nb, emf, ekf)
    end
```

```
Nb=1: MF, 8 Matsubara points: max |G_d - [i $\nu$  -  $\Delta(i\nu)]^{-1}$ | = 4.0e-15; KF G^R, 10 points: 2.2e-12
Nb=2: MF, 8 Matsubara points: max |G_d - [i $\nu$  -  $\Delta(i\nu)]^{-1}$ | = 3.9e-14; KF G^R, 10 points: 3.9e-11
Nb=3: MF, 8 Matsubara points: max |G_d - [i $\nu$  -  $\Delta(i\nu)]^{-1}$ | = 1.1e-14; KF G^R, 10 points: 1.3e-13
```

I1, I2 — PSFs that scale

The dense `psf` visits all n^ℓ eigenstate tuples: $16^4 \approx 6.6 \times 10^4$ per permutation for $N_b = 1$, $64^4 \approx 1.7 \times 10^7$ for $N_b = 2$, and $256^4 \approx 4.3 \times 10^9$ for $N_b = 3$. Eq. (28) needs only the cycles along nonzero matrix elements. The chain walk stores each rotated, `otol`-filtered operator as row-wise adjacency lists and walks $s_1 \rightarrow s_2 \rightarrow s_3 \rightarrow s_4 \rightarrow s_1$ directly, writing straight into the merged `PSF4` storage (I1). Merging uses a `Dict` on positions rounded to `TOL_DEG` cells, followed by a neighbour pass that re-joins a peak split across a cell boundary (I2). The dense `psf` and the pairwise merge are kept as references.

The equality checks first (all 24 permutations, parts `:full` and `:connected`), then the cost table: wall time for all 24 connected tables,

surviving cycles of S per permutation, and merged peaks of S^{con} per permutation.

```
In [6]: function sametabs(A, B)
    key(x) = round(x; digits = 9); dw = 0.0
    for (a, b) in zip(A, B)
        (a.p == b.p && length(a.w) == length(b.w)) || return (false, NaN)
        ia, ib = sortperm(key.(a.pos)), sortperm(key.(b.pos))
        all(maximum(abs.(a.pos[x] .- b.pos[y])) ≤ 1e-12 for (x, y) in zip
        (ia, ib)) || return (false, NaN)
        dw = max(dw, maximum(abs(a.w[x] - b.w[y]) for (x, y) in zip(ia, i
        b)); init = 0.0))
    end
    return (true, dw)
end

at = HubbardAtomModel(2.0, 5.0); ao = operators(at)
dm = HubbardDimerModel(0.7, 2.3, 3.0)
systems = [("atom", spectrum(at), (ao.d_up, ao.dag_up, ao.d_dn, ao.dag_d
n), 0.0),
    ("dimer, k = 0000", spectrum(dm), dimer_operators_4p(operators
(dm), (0, 0, 0, 0), :updn), 1e-12),
    ("AIM Nb = 1", spectrum(aim_model(1; U = 2.0, β = 10.0)), imp4
(aim_model(1; U = 2.0, β = 10.0), operators(aim_model(1; U = 2.0, β = 10.
0)), :updn), 1e-12),
    ("AIM Nb = 2", spectrum(aim_model(2; U = 2.0, β = 10.0)), imp4
(aim_model(2; U = 2.0, β = 10.0), operators(aim_model(2; U = 2.0, β = 10.
0)), :updn), 1e-12)]
println("chain walk vs dense PSF, merged tables, all 24 permutations:")
for (name, sp, Os, otol) in systems, part in (:full, :connected)
    ok, dw = sametabs(psf4_tables(sp, Os; part = part, otol = otol, metho
d = :dense), psf4_tables_chain(sp, Os; part = part, otol = otol)[1])
    @printf("    %-16s %-10s identical peak lists: %s    max |Δw| = %.1e
\n", name, part, ok, dw)
end
println("\nDict merge vs pairwise merge on the dense terms:")
for (name, sp, Os, otol) in systems[1:2]
    same = all(merge_peaks(d.terms) == merge_peaks_pairwise(d.terms) for
part in (:full, :connected)
        for d in permuted_psf4_tables(sp, Os; part = part, otol = otol))
    @printf("    %-16s identical output (positions, weights, order): %s
\n", name, same)
end
```

```
chain walk vs dense PSF, merged tables, all 24 permutations:
atom          full      identical peak lists: true    max |Δw| = 0.0e+00
atom          connected  identical peak lists: true    max |Δw| = 0.0e+00
dimer, k = 0000 full      identical peak lists: true    max |Δw| = 0.0e+00
dimer, k = 0000 connected identical peak lists: true    max |Δw| = 1.1e-16
AIM Nb = 1     full      identical peak lists: true    max |Δw| = 0.0e+00
AIM Nb = 1     connected identical peak lists: true    max |Δw| = 2.8e-17
AIM Nb = 2     full      identical peak lists: true    max |Δw| = 0.0e+00
AIM Nb = 2     connected identical peak lists: true    max |Δw| = 5.6e-17
Dict merge vs pairwise merge on the dense terms:
atom          identical output (positions, weights, order): true
dimer, k = 0000 identical output (positions, weights, order): true
```

```
In [7]: m3 = aim_model(3; U = 2.0, β = 10.0)
push!(systems, ("AIM Nb = 3", spectrum(m3), imp4(m3, operators(m3), :upd
n), 1e-12))
psf4_tables(systems[1][2], systems[1][3]; method = :dense); psf4_tables_c
hain(systems[1][2], systems[1][3]) # compile
println(" system          states dense: 24 tables    chain: 24 tables
cycles of S / permutation    merged peaks of S^con / permutation")
```

```

cost = []
for (name, sp, Os, otol) in systems
    n = length(sp.E)
    td = n ≤ 64 ? (@elapsed psf4_tables(sp, Os; part = :connected, otol = otol, method = :dense)) : NaN
    tc = @elapsed ((tabs, st) = psf4_tables_chain(sp, Os; part = :connected, otol = otol))
    push!(cost, (name = name, n = n, dense = td, chain = tc, cycles = extrema(st.cycles), peaks = extrema(st.peaks), total = sum(st.peaks)))
    @printf(" %-16s %5d  %14s  %14s  %8d - %-8d  %8d - %-8d\n", name, n,
            isnan(td) ? "skipped" : @sprintf("%.3f s", td), @sprintf("%.3f s", tc), st.cycles[1] == 0 ? 0 : minimum(st.cycles), maximum(st.cycles),
            minimum(st.peaks), maximum(st.peaks), sum(st.peaks))
end
println("\n(dense skipped for Nb = 3: 2564 ≈ 4.3×109 tuples per permutation)")

```

system	states	dense: 24 tables	chain: 24 tables	cycles of S / permutation	merged peaks
atom	4	0.001 s	0.016 s	1 – 1	4 –
5 (Σ 104)					
dimer, k = 0000	16	0.018 s	0.000 s	8 – 13	20 –
21 (Σ 488)					
AIM Nb = 1	16	0.025 s	0.002 s	64 – 100	110 –
126 (Σ 2896)					
AIM Nb = 2	64	2.830 s	0.147 s	4046 – 5537	4918 –
5249 (Σ 123068)					
AIM Nb = 3	256	skipped	25.021 s	646304 – 814512	301835 –
346159 (Σ 7915652)					

(dense skipped for Nb = 3: 256⁴ ≈ 4.3×10⁹ tuples per permutation)

Up to $N_b = 2$ the chain walk and the dense PSF produce the same merged peaks, with weights equal to 10^{-16} . The Dict merge reproduces the pairwise one exactly, including the order of the peaks. The chain walk makes $N_b = 3$ possible at all: about $0.7\text{--}0.8 \times 10^6$ surviving cycles and 3×10^5 merged peaks per permutation, in tens of seconds. The dense sum would enumerate 4.3×10^9 tuples per permutation.

B1 — the $U = 0$ null test

$S^{\text{con}} \equiv 0$ and $F \equiv 0$, in the MF and all 16 KF components, both spins, $N_b = 1, 2$. As for V5 on the atom, this tests the PSF split of Eq. (31), not the kernel: at $U = 0$ every connected peak cancels before any kernel is applied.

```

In [8]: for Nb in (1, 2)
    m = aim_model(Nb; U = 0.0, β = 10.0); sp = spectrum(m); ops = operators(m)
    ws = fm = fk = 0.0
    for s in (:updn, :upup)
        T, _ = psf4_tables_chain(sp, imp4(m, ops, s); otol = 1e-12, wtol = 0.0)
        ws = max(ws, maximum(maximum(abs, P.w; init = 0.0) for P in T))
        tm = heatmap_tables(m, s, :MF; sp = sp); tk = heatmap_tables(m, s, :KF; sp = sp)
        fm = max(fm, maximum(abs(Fmf(m, tm, ms)) for ms in FREQS))
        buf = zeros{ComplexF64, 16}
        for (v, vp, ω) in ((0.3, -0.2, 0.0), (0.0, 0.1, 0.0), (-0.7, 0.5,

```

```

0.15))
        vertex_point!(buf, v, vp, w, HeatmapParams(0.0, 10.0, 0.1), t
k); fk = max(fk, maximum(abs, buf))
    end
end
    @printf("Nb=%d, U=0: max |S^con weight| before dropping = %.1e; max
|F|: MF %.1e, KF (16 components) %.1e\n", Nb, ws, fm, fk)
end

```

```

Nb=1, U=0: max |S^con weight| before dropping = 3.9e-16; max |F|: MF 0.0e+00, KF (16 components) 0.0e
+00
Nb=2, U=0: max |S^con weight| before dropping = 1.7e-15; max |F|: MF 0.0e+00, KF (16 components) 0.0e
+00

```

B2 — $V \rightarrow 0$: the two branches of Eq. (46) again

At $V = 0$ the impurity decouples and F must equal Eq. (85) exactly (`vertex_exact`). The bath levels are then degenerate with impurity levels, so terms take the anomalous branch of Eq. (46). At $V \neq 0$ they move to the regular branch with near-vanishing denominators. $U = 2$, $\beta = 5$, both spins, seven frequency points including every δ channel.

```

In [9]: U, β = 2.0, 5.0; atom = HubbardAtomModel(U, β)
B2 = Dict{Int,Vector{Float64}}{()}
for Nb in (1, 2)
    @printf("Nb = %d (bath levels %s)\n      V      max |F - F^atom|
dev / V^2\n", Nb, string(aim_model(Nb; U = U, β = β).ε))
    for V in (0.0, 1e-1, 1e-2, 1e-3)
        m = AIMModel(U, β, aim_model(Nb; U = U, β = β).ε, fill(V, Nb))
        e = maximum(abs(Fmf(m, heatmap_tables(m, s, :MF), ms) - vertex_ex
act(atom, s, MatsubaraFreq.(collect(ms))))
            for s in (:updn, :upup), ms in FREQS)
        push!(get!(B2, Nb, Float64[]), e)
        @printf("      %6s      %.3e      %s\n", V == 0 ? "0" : @sprintf
("%.0e", V), e, V == 0 ? "" : @sprintf("%.1f", e / V^2))
    end
end
m = AIMModel(U, β, [-0.5, 0.5], [1e-4, 1e-4])
e12 = maximum(abs(Fmf(m, heatmap_tables(m, s, :MF), ms) - vertex_exact(at
om, s, MatsubaraFreq.(collect(ms)))) for s in (:updn, :upup), ms in FREQ
S)
e8 = maximum(abs(Fmf(m, heatmap_tables(m, s, :MF; otol = 1e-8), ms) - ver
tex_exact(atom, s, MatsubaraFreq.(collect(ms)))) for s in (:updn, :upup),
ms in FREQS)
sp = spectrum(m)
@printf("\nNb = 2, V = 1e-4: dev/V^2 = %.0f with otol = 1e-12, %.1f with o
tol = 1e-8; smallest level spacing %.1e\n", e12 / 1e-8, e8 / 1e-8,
    minimum(filter(>(1e-12), diff(sp.E))))

```

```

Nb = 1 (bath levels [0.0])
  V      max |F - F^atom|      dev / V^2
  0      8.531e-14
1e-01    2.584e+01            2583.7
1e-02    2.942e-01            2942.0
1e-03    2.946e-03            2946.0
Nb = 2 (bath levels [-0.5, 0.5])
  V      max |F - F^atom|      dev / V^2
  0      1.137e-13
1e-01    3.070e+01            3069.7
1e-02    3.591e-01            3591.1
1e-03    3.596e-03            3595.9

```

At $V = 0$ the AIM reproduces Eq. (85) to 10^{-13} . For $V \neq 0$ the deviation follows a clean V^2 law down to $V = 10^{-3}$ for both N_b , so the regular branch takes over from the anomalous one without a jump.

Where it breaks. For $N_b = 2$ at $V = 10^{-4}$ the V^2 law fails by orders of magnitude. The cause is the one seen in D8 of the dimer. Some levels are split only by $\sim V^2 \approx 3 \times 10^{-8}$, so LAPACK mixes the nearly degenerate eigenvectors at $\varepsilon/\text{gap} \sim 10^{-8}$. Symmetry-forbidden matrix elements of that size pass the 1e-12 filter and add spurious peaks. With otol = 1e-8 the V^2 law returns exactly. The small- V limit of a model with split degeneracies needs either an exact treatment of the degeneracy or a filter above ε/gap .

B3 — weak coupling, Eq. (84)

$F_{\sigma\sigma'} = U\delta_{\sigma,\sigma'} - U^2[\delta_{\sigma,\sigma'}\chi_0(i\omega_{12}) + \delta_{\sigma,\bar{\sigma}'}\chi_0(i\omega_{13}) - \chi_0(i\omega_{14})]$ (p. 16), with the discrete-bath bubble [derived on the sheet]

$\chi_0(i\omega_m) = -\sum_{a,b} r_a r_b [n_F(e_a) - n_F(e_b)]/(i\omega_m + e_a - e_b)$, or $+\beta n_F(e_a)(1 - n_F(e_a))$ at $m = 0$, $e_a = e_b$. Here

$G_0 = [i\nu - \Delta(i\nu)]^{-1} = \sum_a r_a/(i\nu - e_a)$ is the particle-hole symmetric, Hartree-shifted propagator, i.e. the $U = 0$ propagator of this very code. The bubble first against the sheet's brute Matsubara sum, then

$\lim_{U \rightarrow 0} (F - U\delta_{\sigma\sigma'})/U^2$. That limit comes from a three-point Richardson extrapolation from $U = 10^{-2}, \frac{1}{2} \cdot 10^{-2}, \frac{1}{4} \cdot 10^{-2}$, which removes the $O(U)$ and $O(U^2)$ corrections; the two-point version of D7 leaves $O(U^2)$ at the 10^{-4} level. $\beta = 10, 20$ frequency triples including points with $\omega_{12} = 0, \omega_{13} = 0, \omega_{14} = 0$.

```
In [10]: m2 = aim_model(2; U = 0.0, β = 10.0)
@printf("χ₀, Nb = 2, β = 10:  χ₀(0) = %.6f  χ₀(iω₁) = %.6f  χ₀(iω₃) = %.6f\n",
        sheet: 1.893894, 0.140828, 0.028022\n",
        real(chi0_bubble(m2, 0)), real(chi0_bubble(m2, 2)), real(chi0_bubble(m2, 6)))
F20 = vcat(FREQS, [(2a + 1, 2b + 1, 2c + 1, -(2a + 2b + 2c + 3)) for (a, b, c) in ((0, 1, -2), (2, -3, 1), (-1, -1, 4), (3, 0, -2), (1, 2, 2), (-2, 1, 0), (0, -4, 3), (4, -1, -1), (-3, 2, 2), (1, -2, -1), (2, 2, -5), (-1, 3, -3), (5, -2, 1))])
β = 10.0; Us = (1e-2, 5e-3, 2.5e-3)
for Nb in (1, 2)
    worst = 0.0; worst2 = 0.0; big = 0.0
    for s in (:updn, :upup)
        tabs = [heatmap_tables(aim_model(Nb; U = U, β = β), s, :MF) for U in Us]
        m1 = aim_model(Nb; U = 1.0, β = β)
        for ms in F20
            f = [(Fmf(aim_model(Nb; U = U, β = β), t, ms) - (s == :updn ? U : 0.0)) / U^2 for (t, U) in zip(tabs, Us)]
            br = vertex_second_order(m1, s, ms) - (s == :updn ? 1.0 : 0.0)
        end
    end
end
```

```

        worst = max(worst, abs((8f[3] - 6f[2] + f[1]) / 3 - br)); worst2 = max(worst2, abs(2f[2] - f[1] - br)); big = max(big, abs(br))
    end
end
    @printf("Nb=%d: max |lim (F - Uδ)/U² - Eq. (84) bracket|, 20 points × 2 spins: 3-point %.1e, 2-point %.1e (max |bracket| = %.2f)\n",
        Nb, worst, worst2, big)
end

```

χ_0 , Nb = 2, $\beta = 10$: $\chi_0(0) = 1.893894$ $\chi_0(i\omega_1) = 0.140828$ $\chi_0(i\omega_3) = 0.028022$ sheet: 1.893894, 0.140828, 0.028022
 Nb=1: max |lim (F - Uδ)/U² - Eq. (84) bracket|, 20 points × 2 spins: 3-point 8.7e-08, 2-point 3.2e-05 (max |bracket| = 1.18)
 Nb=2: max |lim (F - Uδ)/U² - Eq. (84) bracket|, 20 points × 2 spins: 3-point 1.3e-05, 2-point 2.3e-04 (max |bracket| = 1.89)

B4, B5 — crossing and the Keldysh vertex

B4. $F_{\uparrow\uparrow}(i\omega) = F_{\uparrow\downarrow}(i\omega) - F_{\uparrow\downarrow}(i\omega')$, with $\omega_1 \leftrightarrow \omega_3$ and with $\omega_2 \leftrightarrow \omega_4$.

B5. $G^{1111} = 0$ and $F^{2222} \equiv 0$ (p. 15, a general consequence of $G^{11} = 0$).

There is no closed form for the fully retarded vertex, so H4 is generalised: the left side of $2F^{[\eta]}(\omega) = \tilde{G}(z)/[\tilde{G}(z_1)\tilde{G}(-z_2)\tilde{G}(z_3)\tilde{G}(-z_4)]$, $z = \omega^{[\eta]}$, uses Eq. (67), the (D2) legs and Eq. (76). The right side uses only `regular_sum` (4p and 2p), the regular Matsubara kernel at complex frequencies. Since the MF disconnected part is purely anomalous, $\tilde{G}^{\text{con}} = \tilde{G}$ as rational functions [derived on the sheet].

Trap 1. $F_{\uparrow\uparrow}^{[\eta]} \equiv 0$ holds for the atom only, whose $F_{\uparrow\uparrow}$ is purely anomalous (Eq. 85b). The AIM's has a regular part already at $O(U^2)$, so its retarded components do not vanish.

```

In [11]: for Nb in (1, 2)
    m = aim_model(Nb; U = 2.0, β = 5.0); sp = spectrum(m); ops = operator
    s(m)
    tud = heatmap_tables(m, :updn, :MF; sp = sp); tuu = heatmap_tables(m,
    :upup, :MF; sp = sp)
    c13 = maximum(abs(Fmf(m, tuu, ms) - (Fmf(m, tud, ms) - Fmf(m, tud, (m
    s[3], ms[2], ms[1], ms[4])))) for ms in FREQS)
    c24 = maximum(abs(Fmf(m, tuu, ms) - (Fmf(m, tud, ms) - Fmf(m, tud, (m
    s[1], ms[4], ms[3], ms[2])))) for ms in FREQS)
    γ0 = 0.1; prm = HeatmapParams(2.0, 5.0, γ0)
    du, duu = impurity_operators(m, ops, :up)
    leg2 = permuted_psfs(sp, (du, duu); otol = 1e-12)
    Gt(x) = regular_sum(sp, (du, duu), [x, -x]; cache = leg2)
    buf = zeros(ComplexF64, 16); g = zeros(ComplexF64, 16)
    gl111 = f2222 = rel = 0.0; ret_uu = 0.0
    for s in (:updn, :upup)
        Os = imp4(m, ops, s); full = permuted_psfs(sp, Os; otol = 1e-12)
        tfull = heatmap_tables(m, s, :KF; sp = sp, part = :full); tcon =
        heatmap_tables(m, s, :KF; sp = sp)
        for (v, vp, w) in ((0.3, -0.2, 0.0), (0.0, 0.1, 0.0), (-0.7, 0.5,
        0.15), (0.45, 0.45, -0.3))
            kf_correlator_point!(g, ph_frequencies(v, vp, w), γ0, tfull.c
            on); gl111 = max(gl111, abs(g[1]))
            vertex_point!(buf, v, vp, w, prm, tcon); f2222 = max(f2222, a
            bs(buf[16]) / maximum(abs, buf))
        end
    end
end

```

```

w = collect(ph_frequencies(v, vp, w))
for η in 1:4
    z = omega_eta(w, η, γ0)
    rhs = regular_sum(sp, 0s, z; cache = full) / (Gt(z[1]) *
Gt(-z[2]) * Gt(z[3]) * Gt(-z[4]))
    c = LinearIndices((2, 2, 2, 2))[ntuple(i -> i == η ? 1 :
2, 4)...]
    rel = max(rel, abs(2buf[c] - rhs) / abs(rhs))
    s === :upup && (ret_uu = max(ret_uu, abs(buf[c])))
end
end
end
@printf("Nb=%d: B4 crossing ω1↔ω3 %.1e, ω2↔ω4 %.1e | B5 |G1111| =
%.0e, |F2222|/max|F| = %.0e, Fη[η] vs regular sums: %.1e relative, max|
Fη[η]↑↑| = %.1f\n",
Nb, c13, c24, g1111, f2222, rel, ret_uu)
end

```

Nb=1: B4 crossing ω₁↔ω₃ 1.1e-14, ω₂↔ω₄ 1.4e-14 | B5 |G¹¹¹¹| = 0e+00, |F²²²²|/max|F| = 0e+00, F^η[η] vs regular sums: 7.2e-15 relative, max|F^η[η]_{↑↑}| = 117.8
Nb=2: B4 crossing ω₁↔ω₃ 3.7e-13, ω₂↔ω₄ 3.8e-13 | B5 |G¹¹¹¹| = 0e+00, |F²²²²|/max|F| = 0e+00, F^η[η] vs regular sums: 7.0e-13 relative, max|F^η[η]_{↑↑}| = 45.4

B6 — cost

The production setting of the heat maps: $U = 1$, $T = U/50$, $\gamma_0 = T$, $\Delta = 0.1$. Timed here: building the four tables (MF and KF, both spins), and the warm time of one MF 48×48 and one KF 401×401 slice ($\uparrow\downarrow$, all 16 components), with the merged peaks per permutation that set the cost per point. The warm time is a second call, after the first has compiled everything. The KF slices are reused in O3.

```

In [12]: HU, HT = 1.0, 1 / 50; hprm = HeatmapParams(HU, 1 / HT, HT)
msg = [2n + 1 for n in -24:23]; vkf = collect(range(-1, 1; length = 401))
KFstore = Dict{Any,Array{ComplexF64,6}}{()}
B6 = []
for Nb in (1, 2)
    m = aim_model(Nb; U = HU, β = 1 / HT, Δ = 0.1); sp = spectrum(m)
    ttab = @elapsed (tm = heatmap_tables(m, :updn, :MF; sp = sp); tk = he
atmap_tables(m, :updn, :KF; sp = sp);
    heatmap_tables(m, :upup, :MF; sp = sp); heatmap_tabl
es(m, :upup, :KF; sp = sp))
    vertex_grid(msg, msg, 0, hprm, tm); tmf = @elapsed vertex_grid(msg, m
sg, 0, hprm, tm)
    tkf1 = @elapsed vertex_grid(vkf, vkf, 0.0, hprm, tk)
    tkf2 = @elapsed (KFstore[(Nb, 0.1, :updn)] = vertex_grid(vkf, vkf, 0.
0, hprm, tk))
    pk = [length(P.w) for P in tm.con]
    push!(B6, (Nb = Nb, tables = ttab, mf = tmf, kf1 = tkf1, kf2 = tkf2,
peaks = extrema(pk), total = sum(pk)))
    @printf("Nb=%d: 4 tables %.2f s | merged peaks/permutation %d--%d (Σ %
d) | MF 482 warm %.3f s | KF 4012×16 first %.1f s, warm %.1f s (%d threa
ds)\n",
Nb, ttab, extrema(pk)..., sum(pk), tmf, tkf1, tkf2, Threads.n
threads())
end
println("atom, for comparison (heat-map section of the atom notebook): KF
4012×16 warm 2.2 s on one thread, 84 merged peaks in total")

```

Nb=1: 4 tables 0.14 s | merged peaks/permutation 53–73 (Σ 1556) | MF 48² warm 0.032 s | KF 401²×16 first 9.0 s, warm 9.1 s (4 threads)
Nb=2: 4 tables 0.58 s | merged peaks/permutation 630–744 (Σ 16364) | MF 48² warm 0.315 s | KF 401²×16 first 94.1 s, warm 101.2 s (4 threads)
atom, for comparison (heat-map section of the atom notebook): KF 401²×16 warm 2.2 s on one thread, 84 merged peaks in total

01 — moment screening in the local susceptibility

$\chi_{\text{loc}} = \int_0^\beta d\tau \langle S_d^z(\tau) S_d^z \rangle$ is minus the bosonic $\ell = 2$ correlator of Eq. (39) at $\omega = 0$. The Curie term of degenerate states comes out of the anomalous branch of Eq. (46). $N_b = 1$, $U = 2$, $\Delta = 0.1$, $\Delta_{\text{ST}} = 0.2104$. The sheet's values are given for comparison.

```
In [13]: sheet = [(1.0, 0.18048, 0.18276), (10.0, 1.95387, 2.49989), (30.0, 2.2344
0, 7.5), (100.0, 2.02469, 25.0), (1000.0, 2.02467, 250.0)]
println("      β      χ_loc AIM (Nb=1)      sheet      χ_loc atom      sheet")
for (β, a, t) in sheet
    @printf("    %6.0f      %5f      %5f      %9.5f      %9.5f\n", β, local_susceptibility(aim_model(1; U = 2.0, β = β)), a,
        local_susceptibility(HubbardAtomModel(2.0, β)), t)
end
println("\nthe parity of Nb decides screening (U = 1, Δ = 0.1): half filling holds Nb + 1 electrons")
for Nb in 1:3
    m = aim_model(Nb; U = 1.0, β = 50.0); sp = spectrum(m)
    g = count(e -> e - sp.E[1] < 1e-9, sp.E)
    @printf("    Nb=%d: %d electrons, ground-state degeneracy %d; χ_loc at β = 50, 500: %3f, %3f\n", Nb, Nb + 1, g,
        local_susceptibility(m), local_susceptibility(aim_model(Nb; U = 1.0, β = 500.0)))
end
```

β	χ _{loc} AIM (Nb=1)	sheet	χ _{loc} atom	sheet
1	0.18048	0.18048	0.18276	0.18276
10	1.95387	1.95387	2.49989	2.49989
30	2.23440	2.23440	7.50000	7.50000
100	2.02469	2.02469	25.00000	25.00000
1000	2.02467	2.02467	250.00000	250.00000

the parity of Nb decides screening (U = 1, Δ = 0.1): half filling holds Nb + 1 electrons
Nb=1: 2 electrons, ground-state degeneracy 1; χ_{loc} at β = 50, 500: 1.154, 1.153
Nb=2: 3 electrons, ground-state degeneracy 2; χ_{loc} at β = 50, 500: 10.839, 108.010
Nb=3: 4 electrons, ground-state degeneracy 1; χ_{loc} at β = 50, 500: 2.964, 2.725

The table is reproduced to all digits. The atom's free moment gives a Curie law, $\chi \rightarrow \beta/4$. With one bath site the moment is bound into a singlet and χ_{loc} saturates at about 2.02 once $\beta\Delta_{\text{ST}} \gg 1$.

One point the task sheet does not make: this depends on the **parity of N_b** . At half filling the system holds $N_b + 1$ electrons. For $N_b = 1, 3$ the number is even, the ground state is a singlet and χ_{loc} saturates. For $N_b = 2$ it is odd, the ground state is a Kramers doublet, and $\chi_{\text{loc}} \propto \beta$ keeps growing at any coupling. A finite discrete bath screens the moment only for odd N_b ; `singlet_triplet_gap` refuses even N_b for that reason.

O2 — the $\delta(\omega_{12})$ term against $\beta\Delta_{ST}$

The analysis of D11, repeated for the $N_b = 1$ AIM at three couplings, $\Delta = 0.1, 0.03, 0.01$ ($U = 2$). The quantity is `delta_part`: the anomalous term of Eq. (45) in the ω_{12} channel, on the connected PSF, amputated like the vertex, at $\omega = (\nu_0, -\nu_0, \nu_1, -\nu_1)$. It is shown relative to the atom's free-moment value $\beta u^2 \tanh \prod(i\omega + u) / \prod(i\omega)$ at the same β . A point is dropped when the connected anomalous term is below 10^{-13} of the full one, i.e. at the roundoff floor.

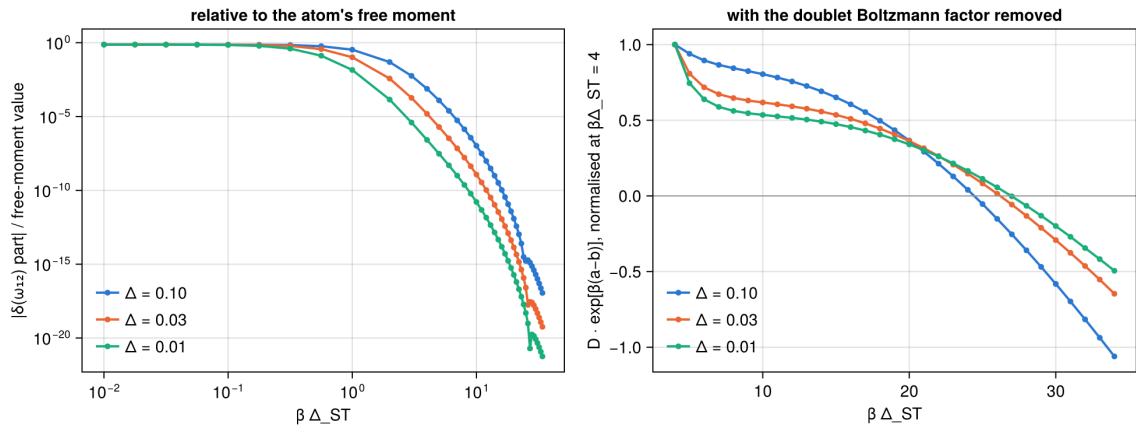
```
In [14]: U = 2.0; ms0 = MatsubaraFreq.([1, -1, 3, -3])
O2 = Dict{Float64,Any}()
for Δ in (0.1, 0.03, 0.01)
    m1 = aim_model(1; U = U, β = 1.0, Δ = Δ); gap = singlet_triplet_gap_n
    b1(m1)
    a, b = sqrt(U^2 + 64m1.V[1]^2) / 4, sqrt(U^2 + 16m1.V[1]^2) / 4
    rows = []
    for x in vcat(10.^ range(-2, 0; length = 9)[1:end-1], 1.0:1.0:34.0)
        β = x / gap; m = aim_model(1; U = U, β = β, Δ = Δ); sp = spectrum
        (m); ops = operators(m)
        Os = imp4(m, ops, :updn)
        ac = abs(anomalous_part(m, Os, ms0; channel = [1, 2], sp = sp, pa
        rt = :connected, otol = 1e-12))
        af = abs(anomalous_part(m, Os, ms0; channel = [1, 2], sp = sp, pa
        rt = :full, otol = 1e-12))
        D = delta_part(m, :updn, ms0; sp = sp, ops = ops)
        u = U / 2; iw = [im * value(f, β) for f in ms0]
        free = abs(β * u^2 * tanh(β * u / 2) * prod(iw .+ u) / prod(iw))
        push!(rows, (x = x, β = β, D = real(D), R = abs(D) / free, ok = a
        c > 1e-13 * af))
    end
    O2[Δ] = (gap = gap, dgap = a - b, rows = rows)
    r = rows
    lr = [(r[i].x + r[i+1].x) / 2 => -(log(abs(r[i+1].D)) - log(abs(r[i].
    D))) / (r[i+1].β - r[i].β) / gap
        for i in eachindex(r)[1:end-1] if r[i].x ≥ 3 && r[i+1].x ≤ 15]
        sc = findfirst(i -> r[i].ok && r[i+1].ok && sign(r[i].D) != sign(r[i+
        1].D), eachindex(r)[1:end-1])
        @printf("Δ=%.2f: Δ_ST = %.4f, doublet gap a-b = %.4f = %.3f Δ_ST; R
        at x = 0.01: %.3f; local rate / Δ_ST at x = 3..15: %s; D changes sign at
        x ≈ %.1f\n",
            Δ, gap, a - b, (a - b) / gap, r[1].R, join([@sprintf("%.2f",
        v) for (_, v) in lr], " "), sc == nothing ? NaN : (r[sc].x + r[sc+1].x)
        / 2)
    end
    jldsave(joinpath(FIGDIR, "o2_delta_part.jld2"); O2 = Dict{string(k) => (g
    ap = v.gap, dgap = v.dgap, x = [r.x for r in v.rows], beta = [r.β for r i
    n v.rows],
        D = [r.D for r in v.rows], R = [r.R for r in v.rows], ok = [r.ok
    for r in v.rows]) for (k, v) in O2))
```

Δ=0.10: Δ_ST = 0.2104, doublet gap a-b = 0.1503 = 0.715 Δ_ST; R at x = 0.01: 0.750; local rate / Δ_ST at x = 3..15: 0.74 0.78 0.76 0.75 0.74 0.74 0.74 0.74 0.75 0.76 0.76 0.77; D changes sign at x ≈ 24.5
 Δ=0.03: Δ_ST = 0.0713, doublet gap a-b = 0.0526 = 0.737 Δ_ST; R at x = 0.01: 0.750; local rate / Δ_ST at x = 3..15: 1.07 0.95 0.86 0.80 0.78 0.76 0.76 0.76 0.76 0.76 0.77 0.78; D changes sign at x ≈ 26.5

$\Delta=0.01$: $\Delta_{ST} = 0.0248$, doublet gap $a-b = 0.0185 = 0.745 \Delta_{ST}$; R at $x = 0.01$: 0.750; local rate / Δ_{ST} at $x = 3...15$: 1.28 1.04 0.90 0.83 0.79 0.77 0.77 0.76 0.76 0.77 0.77 0.78; D changes sign at $x \approx 26$.
5

```
In [15]: palette = Dict{0.1 => "#2a78d6", 0.03 => "#eb6834", 0.01 => "#1baf7a"}
fig = Figure(size = (1000, 390))
ax1 = Axis(fig[1, 1]; xscale = log10, yscale = log10, xlabel = " $\beta \Delta_{ST}$ ",
ylabel = " $|\delta(\omega_{12}) \text{ part}| / \text{free-moment value}$ ",
title = "relative to the atom's free moment")
ax2 = Axis(fig[1, 2]; xlabel = " $\beta \Delta_{ST}$ ", title = "with the doublet Boltzmann factor removed")
for  $\Delta$  in (0.1, 0.03, 0.01)
    r = [x for x in 02[ $\Delta$ ].rows if x.ok]
    lab = @sprintf(" $\Delta = \%.2f$ ",  $\Delta$ )
    scatterlines!(ax1, [x.x for x in r], [x.R for x in r]; color = palette[ $\Delta$ ], markersize = 7, linewidth = 2, label = lab)
    rr = [x for x in r if x.x  $\geq$  4]
    scatterlines!(ax2, [x.x for x in rr], [x.D * exp(x. $\beta$  * 02[ $\Delta$ ].dgap) / abs(rr[1].D * exp(rr[1]. $\beta$  * 02[ $\Delta$ ].dgap)) for x in rr];
    color = palette[ $\Delta$ ], markersize = 7, linewidth = 2, label = lab)
end
hlines!(ax2, [0.0]; color = :gray60, linewidth = 1)
axislegend(ax1; position = :lb, framevisible = false); axislegend(ax2; position = :lb, framevisible = false)
ax2.ylabel = " $D \cdot \exp[\beta(a-b)]$ , normalised at  $\beta \Delta_{ST} = 4$ "
resize_to_layout!(fig)
save(joinpath(FIGDIR, "o2_delta_part.png"), fig; px_per_unit = 2); save(joinpath(FIGDIR, "o2_delta_part.pdf"), fig)
fig
```

Out[15]:



Reading O2. The cut-off is there, but the dimer's law does not carry over.

- **High temperature.** For $\beta \Delta_{ST} \ll 1$ the $\delta(\omega_{12})$ term is 0.75 of the atom's free-moment value, not 1: the bath site takes part in the thermal ensemble.
- **Crossover.** It sits at $\beta \Delta_{ST} \approx 1$, as in the dimer.
- **Beyond it, the decay is not set by Δ_{ST} .** For $\beta \Delta_{ST} \approx 6-15$ the local rate $-d \ln |D| / d\beta$ is 0.74-0.78 Δ_{ST} for all three couplings. That is the **doublet gap** $a - b$, with $(a - b) / \Delta_{ST} = 0.715, 0.737, 0.745$. For the $N_b = 1$ AIM the lowest excitation above the singlet is the four-fold odd-charge level $-U/4 - b$ (A1), which lies *below* the triplet. Degenerate doublets carry a Curie-like $\beta \delta_\omega$ term too. In the dimer the triplet was the lowest excitation.

- **A sign change.** Near $\beta\Delta_{\text{ST}} \approx 24.5\text{--}26.5$ the term changes sign. With $e^{-\beta(a-b)}$ divided out (right panel), what remains is a slowly varying prefactor that crosses zero: two thermally excited contributions of opposite sign, from the doublets and the triplet. Near the node the local rate climbs toward Δ_{ST} , which is why a naive two-point rate between $\beta\Delta_{\text{ST}} = 19$ and 40 would suggest "the same law as the dimer".

So the answer to the task's question: exponential cut-off, yes; $\propto \beta e^{-\beta\Delta_{\text{ST}}}$, no. The rate is the lowest excitation that is degenerate, here $a - b$. This is an observation from the numerics, not a derivation.

O3 — heat maps next to the atom's

P1 (MF, Fig. 10 conventions) and P2 (KF, Fig. 11 conventions) at $U = 1$, $T = U/50$, $\gamma_0 = T$, $\omega = 0$. The rows are:

1. the atom;
2. $N_b = 1$ at $\Delta = 4 \times 10^{-4}$ ($\beta\Delta_{\text{ST}} = 0.10$, moment effectively free);
3. $N_b = 1$ at $\Delta = 0.1$ ($\beta\Delta_{\text{ST}} = 15.7$, moment screened);
4. $N_b = 2$ at the same two couplings.

For $N_b = 2$ there is no singlet-triplet regime (O1: odd electron number, Kramers-doublet ground state), so it is shown at the two couplings as they are. Colour scales, normalisations and `symlog` thresholds are those of the atom's figures.

```
In [16]: cases = [("atom", HubbardAtomModel(HU, 1 / HT)), ("Nb=1, Δ=4e-4", aim_model(1; U = HU, β = 1 / HT, Δ = 4e-4)),
                ("Nb=1, Δ=0.1", aim_model(1; U = HU, β = 1 / HT, Δ = 0.1)), ("Nb=2, Δ=4e-4", aim_model(2; U = HU, β = 1 / HT, Δ = 4e-4)),
                ("Nb=2, Δ=0.1", aim_model(2; U = HU, β = 1 / HT, Δ = 0.1))]
for (lab, m) in cases[2:end]
    g = try; @sprintf("βΔ_ST = %.2f", singlet_triplet_gap(m) / HT); catch; "doublet ground state"; end
    @printf("%-14s V = %s %s\n", lab, string(round.(m.V; sigdigits = 3)), g)
end
supers(e) = replace(string(e), "-" => "-", "0" => "0", "1" => "1", "2" => "2", "3" => "3", "4" => "4", "5" => "5", "6" => "6", "7" => "7", "8" => "8", "9" => "9")
function symlog_ticks(lt, decades)
    vals = vcat([-(10.0^d) for d in decades], 0.0, [10.0^d for d in reverse(decades)])
    (symlog.(vals, lt), [v == 0 ? "0" : (v < 0 ? "-" : "") * "10" * supers(round(Int, log10(abs(v)))) for v in vals])
end
RDBU = Reverse(:RdBu)
nothing
```

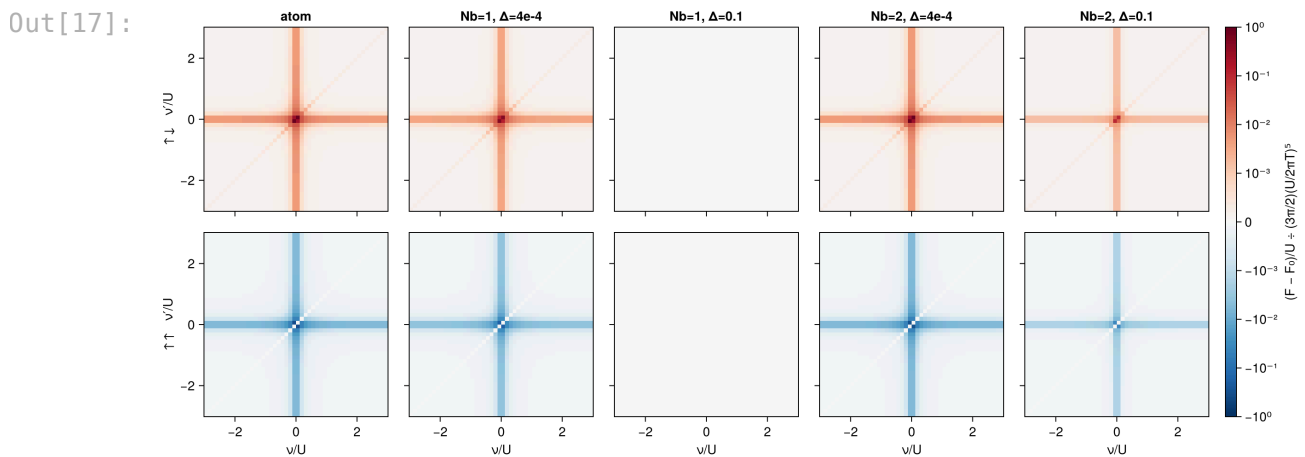
```
Nb=1, Δ=4e-4  V = [0.016]   βΔ_ST = 0.10
Nb=1, Δ=0.1   V = [0.252]   βΔ_ST = 15.66
Nb=2, Δ=4e-4  V = [0.0113, 0.0113] doublet ground state
Nb=2, Δ=0.1   V = [0.178, 0.178] doublet ground state
```

```

In [17]: vmf = msg .* π .* HT ./ HU; normmf = 3π / 2 * (HU / (2π * HT))^5
P1 = Dict{Any,Matrix{Float64}}{()}
for (lab, m) in cases, s in (:updn, :upup)
    F = vertex_grid(msg, msg, 0, hprm, heatmap_tables(m, s, :MF))
    P1[(lab, s)] = real.(F .- (s === :updn ? HU : 0.0)) ./ HU ./ normmf
end
fig = Figure(size = (1300, 560))
for (c, (lab, _)) in enumerate(cases), (r, s) in enumerate(:updn, :upup)
    ax = Axis(fig[r, c]; title = r == 1 ? lab : "", aspect = 1, width = 200, height = 200,
        xlabel = r == 2 ? "v/U" : "", ylabel = c == 1 ? (s === :updn ? "↑↓ v'/U" : "↑↑ v'/U") : "")
    heatmap!(ax, vmf, vmf, symlog.(P1[(lab, s)], 1e-3); colormap = RDBU,
        colrange = (-1, 1), rasterize = 2)
    c > 1 && hideydecorations!(ax; grid = false, ticks = false); r == 1 &&
    & hidexdecorations!(ax; grid = false, ticks = false)
end
Colorbar(fig[1:2, 6]; colormap = RDBU, limits = (-1, 1), ticks = symlog_ticks(1e-3, 0:-1:-3), label = "(F - F₀)/U ÷ (3π/2)(U/2πT)⁵")
resize_to_layout!(fig)
save(joinpath(FIGDIR, "o3_p1_mf.png"), fig; px_per_unit = 2); save(joinpath(FIGDIR, "o3_p1_mf.pdf"), fig)
jldsave(joinpath(FIGDIR, "o3_p1_mf.jld2"); nu = vmf, maps = Dict{String{ASCII}, Matrix{Float64}}{()})
@printf("normalised maxima of |F - F₀| (↑↓): %s\n", join([@sprintf("%s %.2e", lab, maximum(abs, P1[(lab, :updn)])) for (lab, _) in cases], ", "))
fig

```

normalised maxima of $|F - F_0|$ (↑↓): atom 9.51e-01, Nb=1, $\Delta=4e-4$ 4.76e-01, Nb=1, $\Delta=0.1$ 2.17e-05, Nb=2, $\Delta=4e-4$ 9.41e-01, Nb=2, $\Delta=0.1$ 9.76e-02



```

In [18]: normkf = (HU / HT)^3
panels = [((1,2,2,2), real, :updn, "↑↓ 1222 Re"), ((1,2,2,2), imag, :updn, "↑↓ 1222 Im"),
    ((2,1,1,2), imag, :updn, "↑↓ 2112 Im"),
    ((1,1,1,1), imag, :updn, "↑↓ 1111 Im"), ((2,1,1,1), real, :upup, "↑↑ 2111 Re"),
    ((1,2,2,2), real, :upup, "↑↑ 1222 Re"),
    ((1,1,1,1), imag, :upup, "↑↑ 1111 Im")]
tkf = @elapsed for (lab, m) in cases, s in (:updn, :upup)
    key = m isa AIMModel ? (nbath(m), (lab == "Nb=$(nbath(m)), Δ=0.1") ? 0.1 : 4e-4, s) : (0, 0.0, s)
    haskey(KFstore, key) || (KFstore[key] = vertex_grid(vkf, vkf, 0.0, hprm, heatmap_tables(m, s, :KF)))
end
@printf("KF slices for 03: %.0f s\n", tkf)

```

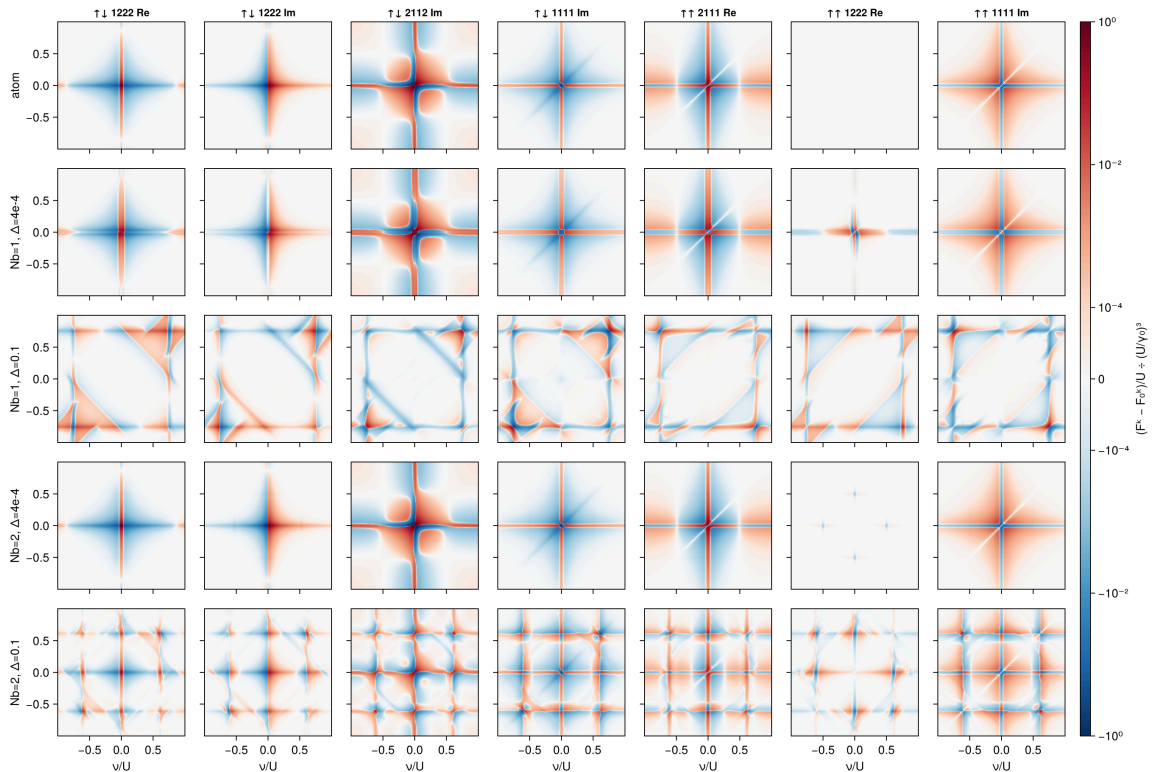
```

keyof(lab, m, s) = m isa AIMModel ? (nbath(m), (lab == "Nb=$(nbath(m)), Δ
=0.1") ? 0.1 : 4e-4, s) : (0, 0.0, s)
fig = Figure(size = (1500, 1100))
for (r, (lab, m)) in enumerate(cases), (c, (k, part, s, title)) in enumer
ate(panels)
    F = KFstore[keyof(lab, m, s)]
    X = part.((F[k..., :, :] .- bare_vertex_kf(k, HU, s == :updn)) ./ HU
    ./ normkf)
    ax = Axis(fig[r, c]; title = r == 1 ? title : "", titlesize = 12, asp
ect = 1, width = 150, height = 150,
        ylabel = c == 1 ? lab : "", xlabel = r == length(cases) ?
        "v/U" : "", xticks = -0.5:0.5:0.5, yticks = -0.5:0.5:0.5)
    heatmap!(ax, vkf, vkf, symlog.(X, 1e-4); colormap = RDBU, colorrange
    = (-1, 1), rasterize = 2)
    c > 1 && hideydecorations!(ax; grid = false, ticks = false, label = t
    rue)
    r < length(cases) && hidexdecorations!(ax; grid = false, ticks = fals
    e)
end
Colorbar(fig[1:length(cases), length(panels) + 1]; colormap = RDBU, limit
s = (-1, 1), ticks = symlog_ticks(1e-4, 0:-2:-4),
    label = "(Fk - F0k)/U ÷ (U/γ0)3")
resize_to_layout!(fig)
save(joinpath(FIGDIR, "o3_p2_kf.png"), fig; px_per_unit = 2); save(joinpa
th(FIGDIR, "o3_p2_kf.pdf"), fig)
fig

```

KF slices for 03: 322 s

Out[18]:



Reading O3. The MF maxima printed above put numbers on what the colours show.

MF (P1).

- $N_b = 1$ with the moment effectively free ($\beta\Delta_{ST} = 0.10$): the atom's pattern, with the cross at $\nu^{(i)} = \pm\pi T$ from the $\beta\delta_\omega$ terms of Eq. (85), at

about half the atom's amplitude (0.48 vs 0.95).

- $N_b = 1$ with the moment screened ($\beta\Delta_{\text{ST}} = 15.7$): the panel is blank, because the whole vertex is 2×10^{-5} of the atom's, below the colour threshold. This is O2's cut-off, seen on the entire plane.
- $N_b = 2$: at weak coupling it is indistinguishable from the atom (0.94). At $\Delta = 0.1$ the cross survives at a tenth of the amplitude (0.098). The Kramers doublet is never screened (O1), so the $\beta\delta_\omega$ terms cannot be cut off.

KF (P2).

- The weakly coupled rows reproduce the atom's panels. Trap 1 is visible directly: the $\uparrow\uparrow$ 1222 column, $F_{\uparrow\uparrow}^{[1]}$, is identically zero for the atom and nonzero for every AIM row.
- The screened $N_b = 1$ row loses the atom's cross at the origin, where the atom has its γ_0 -limited spikes. What remains is structure along lines near $|\nu|, |\nu'| \approx 0.75$, on the scale of the many-body excitation energies.
- $N_b = 2$ at $\Delta = 0.1$ keeps the central cross (unscreened doublet), with replicas along $|\nu|, |\nu'| \approx 0.6$, near the bath energies $\pm D/2$ shifted by the hybridisation.

These readings are visual, not benchmarks.

Summary

Item	Result
Sec. 1	sum rule exact, ε_n symmetric, anticommutators of all $2(N_b + 1)$ modes, $\langle n_{d\sigma} \rangle = \frac{1}{2}$ at every T
I1	chain walk = dense PSF for atom, dimer, $N_b = 1, 2$ (24 permutations, :full / :connected); $N_b = 3$ feasible
I2	Dict merge = pairwise merge, same order; straddled cells re-joined
I3	ImpurityModel ; existing suite unchanged
A1-A3	closed forms $\leq 1\text{e-}13$; the 12 reference values $\leq 4\text{e-}14$; $U = 0$: MF $\leq 4\text{e-}14$, KF $\leq 4\text{e-}11$ (values up to $1/\gamma_0$)
B1	$S^{\text{con}} \leq 2 \times 10^{-15}$, $F \equiv 0$
B2	exact at $V = 0$, $\propto V^2$; eigenvector mixing breaks it at $V = 10^{-4}$ for $N_b = 2$ unless <code>otol</code> $\geq 1\text{e-}8$
B3	Eq. (84) with the discrete-bath χ_0 to $1.3\text{e-}5$ on brackets of order 1-2
B4, B5	crossing $\leq 4\text{e-}13$; $G^{1111} = 0$, $F^{2222} \equiv 0$, $F^{[\eta]}$ = regular sums $\leq 7\text{e-}13$; $F_{\uparrow\uparrow}^{[\eta]} \neq 0$
B6	see the B6 cell
O1	table reproduced; screening only for odd N_b

Item	Result
------	--------

O2	cut-off at the doublet gap $a - b$, with a sign change near $\beta\Delta_{\text{ST}} \approx 25$
----	---